

RETAIL BUSINESS PERFORMANCE AND PROFITABILITY ANALYSIS

Report

PREPARED BY:
Ayushi

Abstract

This project analyzes retail business data to identify factors affecting profitability, inventory management, and store performance. The dataset was collected in CSV format and processed using SQL, Python, and Tableau. SQL was used for data cleaning and profit margin analysis across product categories and sub-categories. Python (Pandas and Matplotlib) was used to analyze the correlation between inventory days and profitability, helping identify the impact of inventory holding on business performance.

Introduction

The project aims to identify profit trends, slow-moving products, and inventory issues in the retail sector. Data analytics techniques were used to generate business insights and improve decision-making.

Tools Used

Tools Used

- SQL – Data cleaning and profit analysis
- Python (Pandas, Matplotlib) – Correlation analysis
- Tableau – Dashboard creation
- CSV Dataset – Data source

Steps Involved

- Downloaded and imported the retail dataset in CSV format.
- Cleaned missing and null values using SQL queries.
- Calculated profit margins by category and sub-category using SQL.
- Used Python to create Inventory Days and Profitability metrics and performed correlation analysis.
- Built Tableau dashboards showing revenue trends, regional performance, profitability, and inventory analysis with filters for region and product type.
- Derived strategic suggestions for slow-moving and overstocked items.

Conclusion

The project successfully demonstrated how SQL, Python, and Tableau can be integrated for retail business analysis. The analysis identified relationships between inventory holding and profitability while highlighting slow-moving and overstocked products. Tableau dashboards helped visualize business performance effectively and provided insights for improving inventory turnover, profitability, and overall decision-making in the retail sector.